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Safe deep reinforcement learning for building energy management

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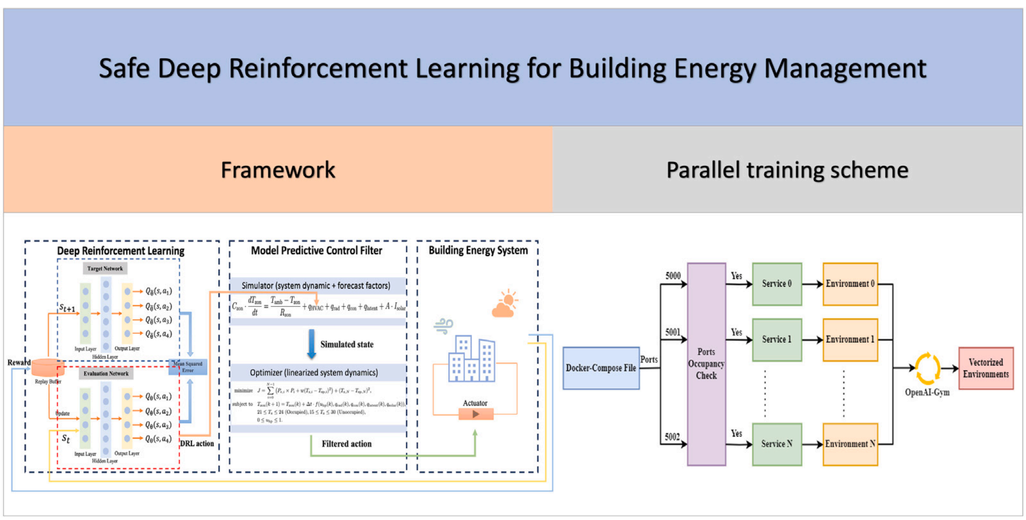
GRAPHICAL ABSTRACT



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ABSTRACT

The optimization of building energy systems poses a complex challenge due to the dynamic nature of building environments and the need for ensuring both energy efficiency and system reliability. This paper proposes a novel approach that synergizes deep reinforcement learning (DRL) with model predictive control (MPC) for safe and efficient building energy management. The approach, named safe DRL, leverages the DRL to explore optimality in uncertain environments and the MPC to dynamically ensure constraints. Off-the-shelf DRL tools are utilized in the safe DRL approach which avoids case-by-case and effort-intensive hyperparameter tuning. MPC with only an approximate building model is then adopted to efficiently filter the DRL action candidates to ensure the safety of the constraints. Additionally, a comprehensive safety metric is proposed considering the duration and severity of constraint violation. Extensive case studies with high-fidelity building models show that the safe DRL approach significantly enhances the constraint satisfaction and improves building performance at the cost of slightly more decision time of a few seconds.

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1. Introduction

Buildings contribute significantly to global energy consumption and carbon emissions. It is estimated that they account for more than 36% of global energy usage and 37% of global carbon dioxide (CO_2) emissions [1]. This fact highlights the pressing need to manage energy consumption in buildings, not only for cost reduction but also for mitigating global climate change. Simultaneously, it is imperative to ensure that energy management strategies do not compromise the occupant comfort and health within buildings because people typically spend approximately 80 to 90 percent of their lives indoors, with nearly two-thirds of this time spent at home [2]. Thus, a delicate balance between energy efficiency and occupant well-being is essential for building energy management, as it not only enhances the sustainability of energy policies but also ensures the well-being and productivity of the inhabitants.

Various advanced energy management and control strategies have been adopted to strike the balance. Among these, three predominant methodologies are frequently employed: rule-based control (RBC) methods, model predictive control (MPC), and control through deep reinforcement learning (DRL) [3]. Each of these control methods offers distinct advantages and challenges, and the choice of them depends on the specific requirements and constraints of the buildings.

Rule-based control (RBC) methods such as proportional–integral–derivative (PID) controller operate on a set of predefined rules or algorithms that trigger specific actions under certain conditions. However, while straightforward, these systems often lack the flexibility to adapt to modern complex or changing environmental conditions [4,5].

On the contrary, MPC represents a more advanced and sophisticated model-based approach, which has been a state-of-art for building control [6]. MPC utilizes a mathematical model of the building and its energy systems to predict future energy needs and optimize control actions accordingly. This predictive capability allows for more efficient energy usage, as the system can anticipate and react to changes in occupancy, weather, and other variables. Furthermore, MPC can handle various control objectives and safety criterion by adjusting the cost function, action and state constraints. It is precisely because of these explicit constraints that MPC usually shows higher security compared with other control strategies [7]. In the context of sophisticated building systems characterized by uncertainties, accurately modeling the thermal dynamics of buildings poses a significant challenge. This limitation can adversely impact the efficacy of MPC. In cases where precise dynamic models are difficult to obtain, some researchers adopt black-box and gray-box models for approximate MPC. Yang et al. [8] utilize a neural network-based black-box MPC to efficiently manage energy in an office building and a lecture hall to avoid constructing explicit dynamic models, achieving energy savings of 58.5% and 36.7%, respectively. Lv et al. [9] develop a gray-box MPC for a thermal zone serviced by a floor heating system and an air-handling unit. They utilize an resistance-capacitance (RC) approach to model the system, with model parameters being updated on a weekly basis. Since the model is obtained by modeling specific objects, the portability remains an issue. Moreover, influenced by environmental uncertainties, initial configurations, and the limited predictive horizon, the actions taken by MPC at each step are not necessarily globally optimal [10].

Contrary to MPC, DRL is an emerging field that leverages the power of artificial intelligence (AI) and machine learning (ML). These systems learn and adapt over time through learning without knowing explicit dynamic models, continually refining their control strategies to maximize efficiency and minimize energy consumption. By designing reward functions in need and processing vast amounts of data, DRL control systems can achieve a higher level of optimization that was previously unattainable by MPC and RBC. Moreover, the trained DRL model can be implemented to different buildings with a small amount of time through online learning [11,12]. Thus, the DRL approach is considered a promising technique in building energy management field. Radaideh and Shirvan [13] initially train agents using proximal policy optimization (PPO) [14] to become proficient in certain problem rules and constraints, after which RL is employed to introduce experiences that assist in steering a range of evolutionary and stochastic algorithms. [15] introduced an approach leveraging the Actor–Critic (A2C) algorithm to enhance the uniformity of radiation temperature across a room through adaptive control of the HVAC system's airflow direction. [16] developed a control strategy for HVAC systems based on the Deep Q-Network (DQN) algorithm, aimed at minimizing energy expenses in office environments while adhering to thermal conditions. [17] presents model-agnostic DRL strategies for residential HVAC optimization, focusing on energy efficiency and occupant comfort using the Deep Deterministic Policy Gradient (DDPG) and DQN algorithms, respectively. [18] proposes a DDPG-based method for managing HVAC systems to simultaneously improve energy usage and thermal comfort in laboratory settings by modulating both temperature and humidity levels. The studies mentioned illustrate that DRL, with its powerful exploration and learning capabilities, is suitable for controlling building energy control, which are nonlinear, time-varying, and uncertain complex systems [11]. However, applying DRL in the real world energy systems raises safety concerns due to its reliance on exploring unknown policies, potentially leading to unpredictable and unsafe behaviors [19,20]. In the building energy management field, if the controller is unsafe, it can result in wasted energy or equipment damage.

To make use of the constraints-satisfactory ability of model-based methods, integrating the constraint-satisfaction capabilities of model-driven approaches with learning-driven methods has been extensively investigated in areas such as robotics, automotive, and video games [21,22]. However, the application of these technologies in the energy community, particularly in building energy management, remains relatively limited. Existing challenges can be categorized into the following areas:

1. **Data Dependence and Computational Costs**: Some approaches rely heavily on substantial data and the high computational cost during the online learning process. Ceusters et al. [23] propose a safety layer based on supervised learning for DRL controllers in multi-energy microgrids. However, this approach relies on extensive training data and learning convergence, which may limit its practicality in dynamic environments. [24] introduces a data-driven MPC (DDMPC) safety filter for RL-based methods targeting photovoltaic battery load systems. The MPC model is trained on state–action pairs derived from RL but it is often challenging to acquire sufficient new data for training. Arroyo et al. [25] propose an algorithm called RL-MPC which combines MPC with DRL. The algorithm is tested on a standardized simulation framework named the building optimization testing (BOPTEST) environment [26], which shows the ability of RL-MPC to meet constraints while possessing the continuous learning property. But due to the absence of an analytical form of the value function, the exhaustive search method is adopted to evaluate all possible actions, which causes concerns on computational efficiency and curse of dimensionality.
2. **Physics and Constraint Handling**: Some articles fail to integrate physical knowledge, relying solely on data-driven methods, which lack theoretical assurance in terms of safety and performance. On the other hand, some articles are overly conservative. The work by Paesschesoone et al. [24] does not adequately utilize physical knowledge, which complicates the theoretical assurance of closed-loop performance. Chen et al. [27] develop an algorithm named Gnu-RL, utilizing differentiable MPC for the end-to-end planning and control specifically designed for HVAC systems. They impose explicit constraints across the entire prediction horizon and tuned cost function through rewards. However, this approach appears conservative due to its implementation of constraints throughout, limiting the operational

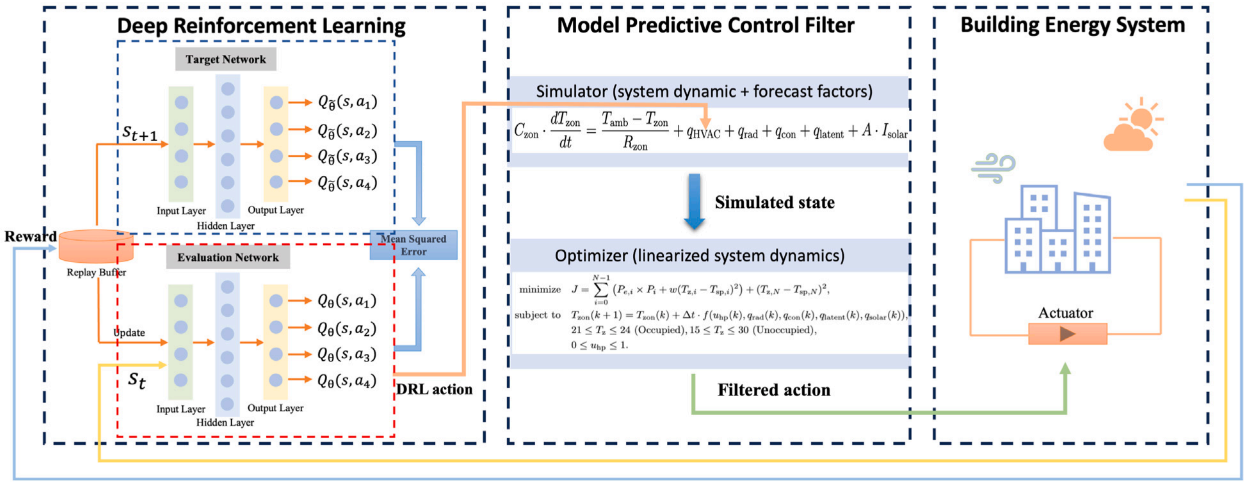


Fig. 1. Safe DRL framework. DRLs in off-the-shelf tools can be applied to avoid case-wise and effort-intensive hyperparameter tuning and approximate model, e.g., a linear one, can be utilized in MPC to facilitate real-time decision making.

space and potential opportunities for performance optimization. Ding et al. [28] model multi-energy system as a constrained Markov Decision Process (CMDP), integrating safety constraints into the objective function as costs. However, this study only considers constant temperature constraints and requires careful design of the penalty terms.

3. **Dimensionality and Practical Implementation**: Some methods cannot handle high-dimensional or continuous state space data, or have not been validated on standard test bench, leading to issues with practical applicability. A hierarchical control is adopted in [29] for HVAC control, where MPC is used for initial planning and Q-learning serves as a real-time controller. But it requires an additional training of a neural network for temperature prediction and the RL method of updating the Q-table through Q-learning is not suitable for high-dimensional state and action spaces. Some researchers have turned to learning-based MPC [30], which employs a statistical model for performance optimization and a deterministic model to enforce comfort constraints. But the method focuses primarily on energy savings rather than thermal comfort, and has not been tested on standard test benches.

In this paper, considering the characteristic of building energy management, a safe DRL framework is proposed to address these issues, where safety is inherently defined by the algorithm's consistent adherence to temperature constraints to ensure thermal comfort. The safe DRL synergizes the learning ability of DRL under uncertainties and the reliability property of physics-based MPC under constraints. DRL is adopted to explore optimal control decisions under uncertainties and MPC is utilized as a safety filter to check online whether the DRL action is safe at each step. DRL action will be rejected and the conservative MPC action will be adopted if unsafety is detected. Thus, an explicit safe set is avoided which contains countless combinations of actions and states for building energy management.

Compared with the existing MPC and DRL algorithms on the standardized testing platform BOPTEST, the proposed algorithm is more stable, secure, and computationally efficient. The main contributions of this paper include:

1. A novel safe DRL framework is proposed, which synergizes the exploratory prowess of DRL in performance optimization with the safety assurance of MPC, dynamically creating an implicit safety set at each step. This leads to marked improvements in key performance indicators (KPIs) embedded in BOPTEST and system safety, underscoring the algorithm's practical effectiveness. The experimental outcomes reveal that safe DRL algorithm can significantly enhance the system's robustness with a maximum decision-making time of up to 2.67 s per step.
2. An open-sourced parallel training scheme is developed for BOPTEST to expedite the training of DRL algorithms for use in the safe DRL. With the parallel training scheme, this study conducts a parallel training of six different DRL algorithms over 10 million steps, followed by an extensive comparative analysis. This approach accelerates the training process, thereby swiftly identifying optimal DRL algorithms suitable for integration into the safe DRL.
3. A safety metric is proposed as an indicator to quantity the safety performance of algorithms during test periods. This metric considers both the severity and duration of unsafety, including measures such as the degree of maximum overshoot or undershoot and the duration of constraint violations.

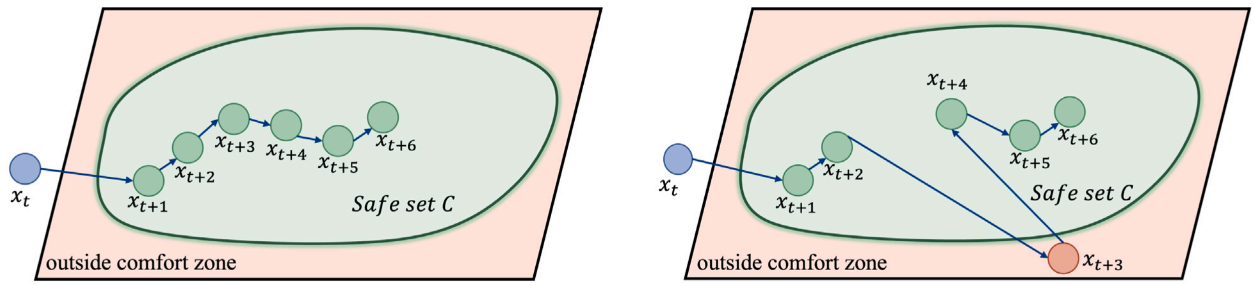
The rest of the paper is organized as follows: Section 2 introduces the safe DRL algorithm and the safety metric. Section 3 presents the test cases of BOPTEST. MPC and safe DRL implementation are shown in Section 4. Section 5 shows the results comparison of different algorithms. Conclusions and future work are provided in Section 6.

2. Safe deep reinforcement learning framework

This section presents the details of the proposed safe DRL framework that synergizes the ability of DRL on optimization exploration and that of MPC on dynamic constraint assurance and an associated safety metric that covers both the duration and severity of constraint violation.

2.1. Safe deep reinforcement learning framework

The safe DRL algorithm is proposed as a synergetic strategy combining MPC with DRL in this subsection. The safe DRL maintains the exploratory benefits of DRL in performance optimization with uncertainties while leveraging the MPC as a safety filter to implicitly create a safety set at every step online. The overall scheme is shown in Fig. 1. Unlike previous safety filters that aimed to minimize the deviation from the actions suggested by RL algorithms [31,32], our MPC filter prioritizes economic efficiency and comfort due to the distinct control scenarios in building energy management, where longer control steps justify increased computational complexity. Additionally, we do not presume that RL-proposed actions are inherently superior. It is unnecessary to adhere strictly to actions with minimal deviations from those



(a) RL actions adopted

(b) MPC actions adopted

Fig. 2. Safe set definition: (a) RL actions adopted (the predicted future states are within the safe set); (b) MPC actions adopted (the predicted future state x_{t+3} exceeds the safe set).

suggested by RL, especially since some unsafe RL actions could pose significant risks. In such cases, it is imperative to distance as much as possible from such actions, as Section 5.2 will show that reliance solely on RL control can lead to detrimental outcomes. The DRL part enables the system to learn optimal behaviors through trial-and-error interactions with the environment and the predictive power of MPC can assess and ensure the dynamic safety of operations proposed by the DRL agent at each step and filter out unsafe actions. Consequently, the safe DRL enhances both the safety, i.e., constraint satisfaction, and performance optimization, e.g., reducing cost, at the cost of negligible overhead of computational time

Remark 1. Selecting hyperparameters in DRL typically demands considerable effort and may necessitate repeating the entire process when the environment, e.g., buildings in this study, undergoes changes. In MPC, attaining dynamics that strike a balance between accuracy and usability can be challenging. However, in the safe DRL framework, off-the-shelf DRL tools can be employed without the need for hyperparameter tuning, while an approximate dynamic model, such as a linear one, can be utilized in the MPC component.

DRL performance optimizer. As the building thermal dynamics exhibit Markovian behavior, i.e., their current states can be completely determined by the states, actions, and other factors at the previous step, we adopt DRL based on Markov decision process (MDP) in the proposed safe DRL framework. An MDP-based DRL for building energy management is characterized by a building energy state space S, an action space A, a reward function $r(s,a) : S \times A \to \mathbb{R}$, and a transition probability function $P(\cdot \mid s,a) : S \times A \to \Delta(S)$, which maps a state–action pair $(s,a) \in S \times A$ to a distribution over the state space. The state space S and action space A can be either discrete or continuous. In the safe DRL for buildings, state transitions are jointly determined by internal factors, e.g., occupancy and thermal mass, and external ones, e.g., ambient temperature and solar irradiation, of the buildings. Deep learning is incorporated to handle high-dimensional state and action spaces. In the DRL part of the safe DRL algorithm, the goal is to find an optimal policy π^* for the building energy management with the consideration of occupant comfort. There are two main approaches: value-based methods by estimating state or state–action values for optimal policy derivation and policy-based methods by directly learning strategies for action selection in each state. Both approaches may be applied in the safe DRL framework and their performance is comprehensively compared in Section 4.2.

MPC safety filter. Safety constraints are considered in the DRL part, but they may be violated due to the exploratory nature of DRL. Thus, we adopt the concept of MPC to dynamically filter the unsafe actions from DRL in real time. The MPC consists of an optimization objective, approximate system dynamics, safety constraints, and other constraints as formulated in the set of Eqs. (1) below.

$$\text{minimize } J = \sum_{i=k}^{k+N-1} [x(i)^T A x(i) + u(i)^T R u(i)] + x(k+N)^T B x(k+N), \tag{1a}$$

$$\text{subject to } x(k+1) = f(x(k), u(k)), \tag{1b}$$

$$u_{\min} \le u(i) \le u_{\max}, \tag{1c}$$

$$x_{\min} \le x(i) \le x_{\max}, \tag{1d}$$

where $x(k)$ denotes the state vector at the kth time step and $u(k)$ is the control input. The cost function in (1a) delineates the optimization criterion, emphasizing the balance between achieving desired temperature trajectories and minimizing energy cost over a prediction horizon N, with A, R and B as weighting matrices. The dynamics of the system is expressed in (1b), which describes how the state of the system evolves over time in response to control actions. Essential to the safety of the safe DRL framework are the action and state constraints in (1c) and (1d), respectively, which ensure that the system operates within predefined operational limits. The MPC implements explicit optimization over a predetermined finite prediction horizon as in (1) and its structure includes not only the optimizer but also a simulator. Note that an approximate dynamic model, e.g., a linear one for building thermal dynamics, is adequate in the safe DRL framework. This ensures the decision efficiency of the framework.

With the proposed safe DRL algorithm, at each control step, the well-trained DRL agent selects an action based on the current state, denoted as $u_{DRL} = \pi^*_\theta(s)$. Concurrently, within the prediction horizon n, the MPC safety filter is employed to record the state at each step, represented as $\{x_{t+1}, x_{t+2}, \ldots, x_{t+N}\}$. The process concludes with a safety constraint check: (1) if $\forall i \in \mathbb{N}, x(k+i) \in C$, where C is the safety set, u_{DRL} from the DRL optimizer is applied; (2) else, an MPC is performed and its optimized control is applied.

As shown in Fig. 2, the safe set is defined as the set of states within a predefined temperature constraint range. If starting from the current RL action, all states within a given finite time horizon remain within the safe set, RL action is utilized. If any state is found to be unsafe, MPC action is employed.

The pseudo code for the safe DRL algorithm is presented in Algorithm 1. First, the MPC and the trained DRL controller are initialized with the same control step settings. During each time step of the test day, the algorithm performs the followings in a loop. The safe DRL first obtain the current state from the building. The DRL controller predicts an action based on the current observation and the MPC simulator initially updates the system state using DRL-predicted action. Subsequently, it integrates MPC optimizer to conduct a forward-looking simulation of the future state trajectories within a defined predictive range. Safety is then determined based on whether the predicted state trajectories surpass the predefined constraint boundaries. The role of the MPC is to ensure that the chosen action does not lead to any unsafe

Algorithm 1: Safe DRL Algorithm

```
Initialize MPC and DRL controllers;
while time in Test Days do
    /* Get current state from the building */
    state ← get_current_state();
    /* Predict action using RL */
    action_rl ← rl_agent.predict(state);
    /* Use MPC to predict the system behavior */
    safe, mpc_action ← mpc.predict(state, action_rl);
    if safe then
        apply_action(action_rl);
    else
        apply_action(mpc_action);
    end
    /* Step the environment to the next state */
    step_environment();
end
```

system states. If the MPC determines that the DRL action is safe, we will execute that action. Otherwise, we will reject the DRL action and employ the action provided by the MPC.

2.2. Safety metric

A safety metric that considers both the duration and severity of the safety violation is designed for quantitative comparison of various algorithms in this subsection. This index consists of two components:

(1) the unsafe time ratio, representing the ratio of the period during which the safety constraint is violated. It is defined as

$$r_{\text{time}} = \frac{t_{\text{violation}}}{t_{\text{total}}}, \tag{2}$$

where r_{time} represents the unsafe time ratio, $t_{\text{violation}}$ the period during which the safety constraint is violated, and t_{total} the total time of the process.

(2) the severity ratio, characterizing the deviation of the unsafe state and the norm state, normalized by the magnitude of the latter. It is defined as

$$r_{\text{sev}} = \beta D(x, C), \tag{3}$$

where r_{sev} is the severity ratio, $D(x, C)$ the distance, e.g., norms in vector space, from the state x to the safety constraint C, and β a scaling factor which may be related to the size of C, the relative magnitude between r_{sev} and r_{time}, etc.

For buildings, the unsafe time ratio is defined as the percentage of the zone temperature overshoot and undershoot time relative to the total testing period, as formulated in (4).

$$r_{\text{time}} = \frac{t_{\text{overshoot}} + t_{\text{undershoot}}}{t_{\text{total}}}, \tag{4}$$

where r_{time} represents the unsafe time ratio, $t_{\text{overshoot}}$ and $t_{\text{undershoot}}$ refer to the total time during which the temperature exceeds its upper limit (overshoot) and falls below its lower limit (undershoot), respectively, and t_{total} represents the total time of the test. Thus, the unsafe time ratio r_{time} represents the percentage of the total test time that the temperature is outside the predefined comfort zone. A lower ratio indicates a better performance of the system in maintaining the temperature within the predetermined safety limits.

The severity ratio is defined as the ratio of the difference between the zone temperature and the temperature upper (lower) limit relative to the temperature upper (lower) bound as formulated in (5).

$$r_{\text{sev}} = \max(P_{\text{overshoot}}, P_{\text{undershoot}}), \tag{5a}$$

$$P_{\text{overshoot}}(t) = \frac{T_{\text{z}}(t) - T_{\text{upper}}(t)}{T_{\text{upper}}(t)}, \tag{5b}$$

$$P_{\text{undershoot}}(t) = \frac{T_{\text{lower}}(t) - T_{\text{z}}(t)}{T_{\text{lower}}(t)}, \tag{5c}$$

where r_{sev} is the severity ratio that represents the maximum of the overshoot and undershoot during the entire testing period, $P_{\text{overshoot}}(t)$ represents the percentage by which the temperature exceeds the upper bound at a specific moment t, T_{z} is the zone temperature value at time t, $T_{\text{upper}}(t)$ is the upper bound limit at time t, and $P_{\text{undershoot}}(t)$ indicates the percentage by which the temperature falls below the lower limit at a specific moment t.

By integrating both the unsafe time ratio and the severity ratio, the proposed safety metric in (6) offers a comprehensive assessment of the performance of the system.

$$m_s = r_{\text{time}} + r_{\text{sev}}, \tag{6}$$

where m_s is the safety metric. The unsafe time ratio reflects the system's stability throughout the entire testing period, while the severity ratio reveals the system's capabilities in the most adverse scenarios.

3. Test case

In this research, to focus on the building energy management task, we utilize the BESTEST Hydronic Heat Pump test case [26], which simulates a single-zone dwelling in Belgium with a diverse set of parameters, including dynamic models, baseline controllers, meteorological data, electrical pricing structures, and carbon emission coefficients.

The test case employed in this paper is inhabited by a five-member family, features a 12 × 16 m floor plan, 2.7 m high ceilings, and south-facing windows. During winter, its heating demand approximates $80\,\text{W/m}^2$. In addition, the BOPTEST environment provides various measurement and control variables: U for control signal, P for measured electric power, Q for thermal power, and T_z for the zone operative temperature. Occupancy is before 7 am and after 8 pm on weekdays, and throughout weekends. The requirements of thermal comfort are that the temperature stays between 21–24 °C when occupied, and 15–30 °C otherwise. Additionally, the emulator features two testing periods scenarios: the peak heat days (January 17th to 31th) and typical heat days (April 19th to May 3rd). There are three electricity pricing scenarios: constant, dynamic and highly dynamics. The detail configuration of the above scenarios is shown in Table 1.

The heating system comprises a 15 kW nominal thermal capacity air-to-water modulating heat pump with 3.5 kW power consumption, connected to a floor heating system. A well-tuned proportional–integral (PI) controller, representing RBC, is included. It initializes the system during warm-up and acts as a benchmark against any implemented controllers.

Table 1
BOPTEST scenarios details.

Scenarios	Categories	Features
Testing periods	Typical heat day	January 17th–31st
	Peak heat day	April 19th–May 3rd
Pricing scenarios	Constant	Fixed rate: 0.2535 €/kWh
	Dynamic	Peak: 0.2666 €/kWh, Off-peak: 0.2383 €/kWh
	Highly dynamic	Based on BELPEX 2019 day-ahead price [33]

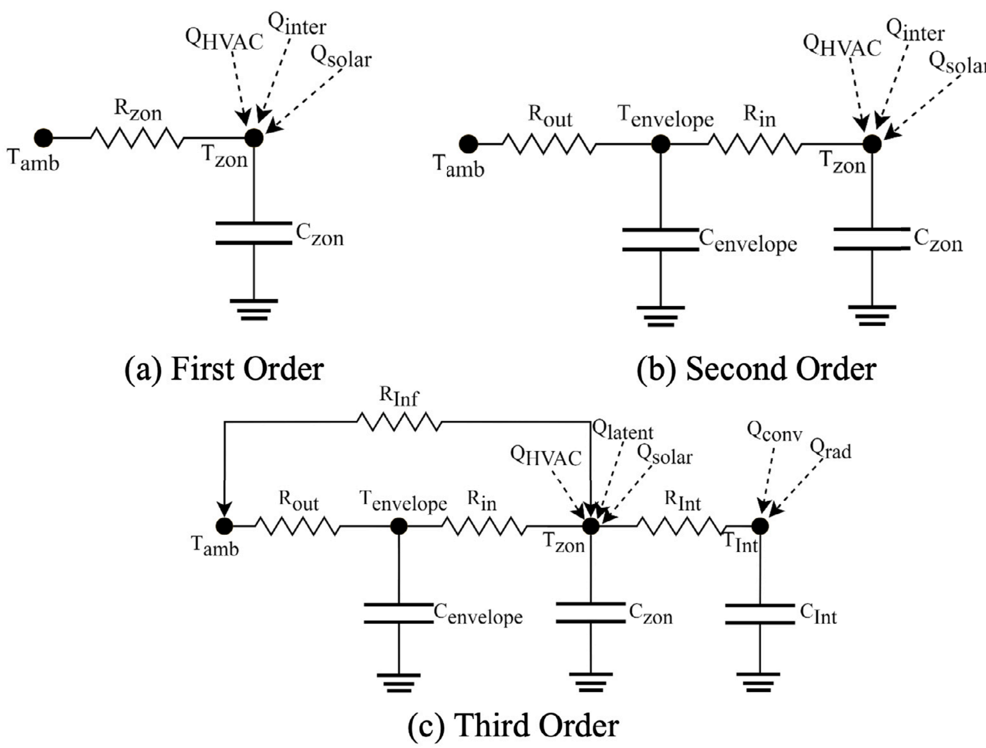


Fig. 3. RC model of different orders.

4. MPC and DRL implementation

This section delineates the implementation specifics of safe DRL algorithm, including the DRL performance optimizer and the MPC safety filter, applied to the test case described in Section 3.

4.1. Model predictive control implementation

MPC implementation includes the system identification, problem formulation, and hyperparameter tuning.

4.1.1. System identification

The performance of MPC is significantly influenced by the system dynamics. To effectively implement MPC in the BESTEST Hydronic Heat Pump test case, constructing a dynamic model of the system is essential. In this study, we develop a data-driven gray-box model for buildings. This subsection begins with a first-order resistance-capacitance (RC) model and uses a forward selection method to gradually increase its complexity with Nelder–Mead method applied for fitting optimization.

The RC model is frequently employed in architectural energy simulation to assess a building's energy efficiency and its indoor thermal environment [34]. In the RC model, resistance represents the impediment of building materials to heat flow. A higher resistance value indicates better insulation, meaning that heat is more difficult to pass through the material. Capacitance signifies the thermal storage capacity of building materials. A higher capacitance value suggests that the material can absorb and store more heat, thereby affecting the rate of change in indoor temperature.

The complexity of the RC model increases with the number of RC combinations. Fig. 3 illustrates the structures of RC models ranging from the first-order to the third-order. Specifically, a first-order model is formulated in (7), a second-order model in (8), which considers the influence of the building envelope, and a third-order model in (9), which further considers building thermal mass.

$$C_{\text{zon}} \cdot \frac{dT_{\text{zon}}}{dt} = \frac{T_{\text{amb}} - T_{\text{zon}}}{R_{\text{zon}}} + q_{\text{HVAC}} + q_{\text{rad}} + q_{\text{con}} + q_{\text{latent}} + A \cdot I_{\text{solar}}, \quad (7)$$

where T_{zon} is the temperature inside the room, T_{amb} the temperature of the ambient environment, R_{zon} the thermal resistance between the environment and the interior of the room, C_{zon} the heat capacity inside the room, q_{HVAC} the heat influx generated by the heating system (unit: W), q_{rad} the general radiation amount, q_{con} the convective heat exchange occurring indoors, q_{latent} the latent heat within the indoor environment, I_{solar} the solar radiation intensity, and A the effective area of the windows.

$$C_{\text{wall}} \cdot \frac{dT_{\text{wall}}}{dt} = \frac{T_{\text{amb}} - T_{\text{wall}}}{R_{\text{out}}} - \frac{T_{\text{wall}} - T_{\text{zon}}}{R_{\text{in}}}, \quad (8a)$$

$$C_{\text{zon}} \cdot \frac{dT_{\text{zon}}}{dt} = \frac{T_{\text{wall}} - T_{\text{zon}}}{R_{\text{in}}} + q_{\text{HVAC}} + q_{\text{rad}} + q_{\text{con}} + q_{\text{latent}} + A \cdot I_{\text{solar}}, \quad (8b)$$

where C_{wall} represents the thermal capacity of the envelope, T_{wall} the temperature of the envelope, R_{in} the thermal resistance between the

Fig. 4. Prediction performance of first-order RC model during the peak heat test period.

Table 2
Training and test MSEs of RC models of various orders.

Order	Training MSE	Test MSE
1	2.37e−09	2.02e−09
2	9.33e−10	3.87e−09
3	9.34e−10	3.79e−09

wall and the interior of the room, and R_{out} the thermal resistance between the wall and the ambient.

$$C_{wall} \cdot \frac{dT_{wall}}{dt} = \frac{T_{amb} - T_{wall}}{R_{out}} - \frac{T_{wall} - T_{zon}}{R_{in}}, \tag{9a}$$

$$C_{int} \cdot \frac{dT_{int}}{dt} = \frac{T_{zon} - T_{int}}{R_{int}} + q_{rad}, \tag{9b}$$

$$C_{zon} \cdot \frac{dT_{zon}}{dt} = \frac{T_{wall} - T_{zon}}{R_{in}} + \left(\frac{T_{amb} - T_{zon}}{R_{Inf}} - \frac{T_{zon} - T_{int}}{R_{int}}\right) + q_{HVAC} + q_{con} + q_{latent} + A \cdot I_{solar}, \tag{9c}$$

where R_{Inf} represents the thermal resistance between the external environment and the room's interior, R_{int} the thermal resistance between the building's thermal mass and the room's interior, T_{int} the temperature associated with the building's thermal mass, and C_{int} the thermal capacity correlated with the building's thermal mass.

In our study, we focus on fitting the system dynamics of the building, as this is crucial for capturing the short-term responses of the system, which are essential for ensuring safe control operations. Instead of modeling the temperature values, we choose to fit the rate of change of the temperature. This method enhances our ability to capture the dynamic nature of temperature fluctuations over time, a key aspect for the predictive control strategies utilized in MPC. By modeling the rate of change in temperature, the MPC can effectively make dynamic adjustments in response to real-time changes in the system. We initiate our identification with a first-order model, employing a forward selection method to incrementally increase the model's order. The fitting optimization algorithm we use is the Nelder–Mead method, a robust optimization algorithm that is insensitive to initial guesses [35]. Subsequently, we compare the mean squared error (MSE) of models in different orders on both the two-week long training and testing data sets generated by the BOPTEST emulator. The results are illustrated in Table 2. The first-order model, with a training MSE of 2.37e−09 and a test MSE of 2.02e−09, demonstrates a good balance between training and testing performance. In contrast, the second and third-order models suggest potential over-fitting issues while they have lower training MSE. Consequently, this study adopts the first-order RC model, the performance of which during the peak heat test period is illustrated in Fig. 4.

Remark 2. The first-order model is selected to balance the training and testing performance based on the data from the emulator, but the selection is case-specific. For other applications, the most appropriate model should be explored and identified via the application-specific data.

4.1.2. *Heating power identification*

The control system regulates the heating power provided to the thermal zone through heat pump control signals. This section utilizes data from the emulator to derive an equation correlating these control signals with the heating output. A higher order signifies greater precision in fitting, yet it also implies increased computational burden in real-time control. After comparison, this article adopts a second-order equation as shown in Eq. (10).

$$q_{HVAC} = (-9510.7 \cdot u_{hp}^2 + 16760 \cdot u_{hp} + 493.7). \tag{10}$$

Additionally, the fitting performance of the model is illustrated in Fig. 5. The fitting results appear to be quite satisfactory, although they do not entirely reflect the true values. For instance, the presence of a constant term results in a non-zero predicted output when the input is zero. However, considering the small margin of error and the fact that the heat pump is typically in operation during most of the heating period, this fitted model will be used for subsequent implementation of the MPC in this paper. In real-world applications, heat flux meters can be employed to measure the heat flux emitted by the HVAC system into a designated zone to derive the relationship between the control signal and the heat flux similar to Eq. (10).

4.1.3. *MPC formulation*

With the system identification, we present the details of the implementation of the MPC safety filter in the safe DRL framework in this subsection.

Specifically, the mathematical formulations of the implemented MPC safety filter is

$$\text{minimize} \quad J = \sum_{i=0}^{N-1} \left(P_{e,i} \times P_i + w(T_{z,i} - T_{sp,i})^2\right) + (T_{z,N} - T_{sp,N})^2, \tag{11a}$$

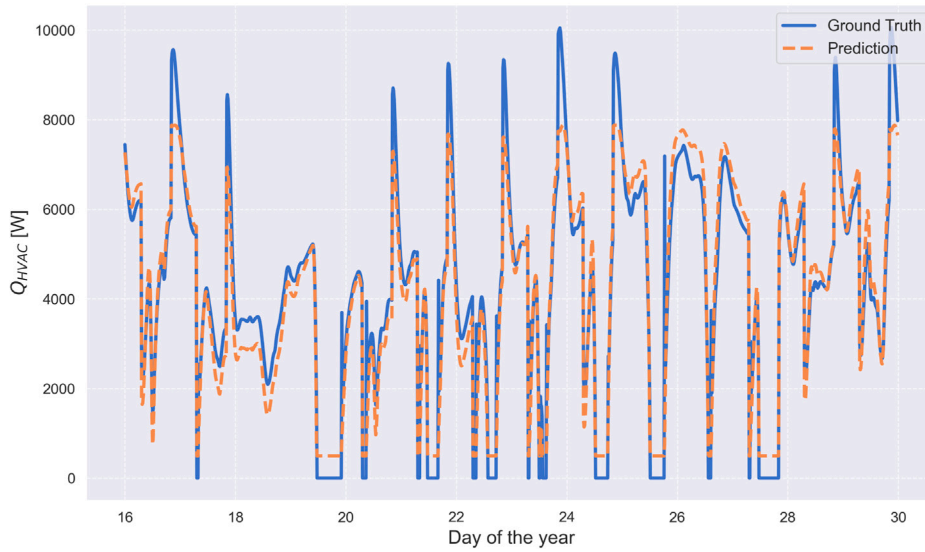


Fig. 5. Heating power identification performance.

$$\text{subject to} \quad T_{\text{zon}}(k+1) = T_{\text{zon}}(k) + \Delta t \cdot f(u_{\text{hp}}(k), q_{\text{rad}}(k), q_{\text{con}}(k), q_{\text{latent}}(k), q_{\text{solar}}(k)), \tag{11b}$$

$$21 \le T_{\text{z}} \le 24 \text{ (Occupied)}, 15 \le T_{\text{z}} \le 30 \text{ (Unoccupied)}, \tag{11c}$$

$$0 \le u_{\text{hp}} \le 1. \tag{11d}$$

As shown in (11a), the cost function comprises stage cost and terminal cost, where the stage cost measures the energy cost and the distance between the operative temperature and the desired trajectory throughout the prediction horizon, and the terminal cost focuses on the discrepancy of the temperature deviation at the end of the prediction horizon, ensuring long-term stability and performance. The primary aim of the MPC controller is twofold: firstly, to reduce energy expenditure, and secondly, to minimize the variance of the zone operative temperature from its predetermined set point. The trade-off between the two objectives is modulated by the parameter w. The first component of the stage cost calculates energy costs by multiplying the electricity usage, P, with the time of use electricity rate, P_e(€/kWh). The second component represents thermal discomfort, which is the deviation of the working area temperature from the set point, T_{sp}, which is set as the mean of the lower and upper temperature bounds.

The dynamics in (11b) represent the discretized version of the first-order RC building dynamic model identified in Section 4.1, as required for the implementation of MPC. The linearity of the model simplifies the optimization problem in the MPC and increases the computational efficiency in the safety filtering process.

The operative temperature of the zone, denoted as T_z, must be maintained within a specific temperature range: between 21 °C and 24 °C during occupied periods, and between 15 °C and 30 °C at other times, as specified in constraint (11c). In addition, the control signal for the heat pump, u_{hp}, is regulated to stay within the bounds of 0 and 1 as in (11d), which corresponds to non-operation and full power operation, respectively.

4.1.4. Hyperparameter tuning

The performance of the MPC safety filter heavily depends on hyperparameters, e.g., the prediction horizon, the control period and the weight. The prediction horizon defines the time span over which the controller anticipates future events and the control period is the interval at which the control algorithm updates its decisions. The weight serves to balance energy consumption costs with thermal discomfort. A longer prediction horizon or a shorter control period can provide more information but increases the computational load. Focusing on total control time (the cumulative duration required for the MPC algorithm to process and respond throughout the control cycle), energy cost (product of electricity cost and consumption) and thermal discomfort (defined as the deviations of zone temperature T_z from the comfort range), Table 3 demonstrates the experimental outcomes of MPC under various combinations of prediction horizons and control steps during peak heat day.

The data from this study suggests that with a 3-h prediction horizon and a 15-min control step, the MPC reaches a reasonable balance in its overall performance. This configuration demonstrates a competent performance in the total control time, taking 145.680 s, which reflects the system's efficiency in making timely and effective control decisions. While the energy cost of 0.938 (€/m^2) may not be the lowest, it offers a satisfactory cost-performance balance. In the aspect of thermal comfort, this setup scores 3.967 (Kh/zone), showing an improved performance relative to other alternatives and contributing positively to indoor environmental comfort. The results indicate that extending the prediction horizon leads to an increase in thermal discomfort, which may stem from the extended prediction horizon complicating the convergence of the MPC optimizer. Consequently, this study selects the MPC configuration with a 3-h prediction horizon and a 15-min control step for a compromise of time efficiency, energy optimization, and thermal comfort.

In Eq. (11a), w represents the hyperparameter of the weight that requires tuning. We conduct multiple experiments testing different values of w under a configuration with a 3-h prediction horizon and a 15-min control step, as illustrated in Fig. 6. Analyzing the data, a trend is observed in both energy cost and thermal discomfort across different values of w. Particularly, as w increases from 0 to 1000, energy cost rises from 0.837 (€/m^2) to 0.976 (€/m^2), indicating a moderate increase in energy expenditure with higher w values. However, the most significant change is seen in thermal discomfort, which drastically drops from a high initial value of 41.963 (Kh/zone) to 1.655 (Kh/zone). This indicates a substantial improvement in thermal comfort. Upon further increasing w to 5000, both energy cost and thermal discomfort exhibit relatively stable trends. Energy cost fluctuates between 0.938 (€/m^2) and 0.985 (€/m^2), while thermal discomfort varies within the range of 1.655 (Kh/zone) to 3.967 (Kh/zone). This suggests that beyond $w = 1000$, further increases in w have limited effects on reducing thermal discomfort, while slightly increasing the energy cost.

Considering both the energy cost and thermal discomfort metrics, selecting $w = 1000$ appears to be a balanced decision. At this point, thermal discomfort is significantly reduced, and energy cost remains

Table 3
Performance metrics of MPC under various prediction horizons and control steps.

Prediction horizon (h)	Control step (min)	Total control time (s)	Cost total (€/m^2)	Thermal discomfort (Kh/zone)
3	15	145.680	0.938	3.967
3	30	74.675	0.972	16.749
3	60	41.507	1.033	158.277
6	15	184.163	0.854	5.334
6	30	82.518	0.959	48.655
6	60	48.555	1.046	187.296
12	15	281.765	0.894	49.377
12	30	106.3501	0.992	144.412
12	60	52.683	1.069	261.952
24	15	460.941	1.097	468.938
24	30	170.510	1.065	407.034
24	60	64.048	1.107	479.710

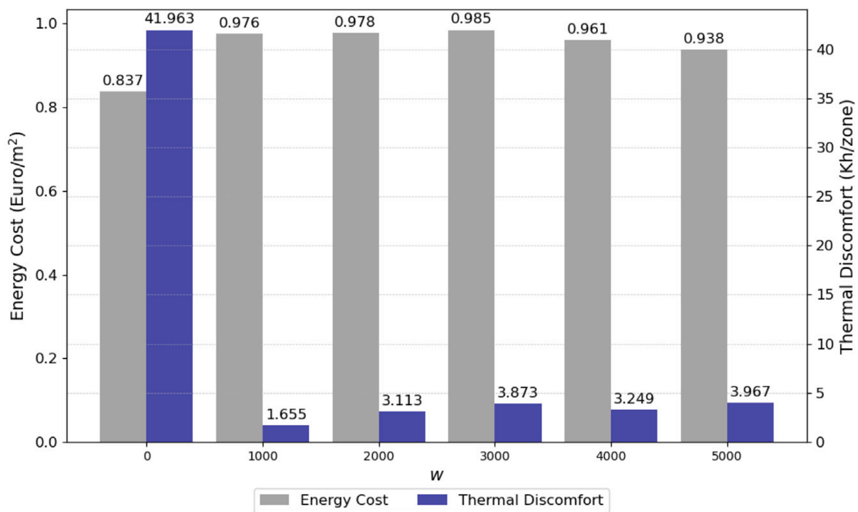


Fig. 6. Test Results under various weight factors w.

within an acceptable range. This balance represents an optimal choice for ensuring comfort while also taking into account energy efficiency and cost-effectiveness. Thus, this paper chooses $w = 1000$ as the optimal parameter.

4.2. Deep reinforcement learning implementation

The implementation of the DRL performance optimizer is detailed. The parallel training scheme is proposed to expedite the training of DRL algorithms using the stable baselines3 tools [36]. Moreover, the reward function, observation and action spaces are designed in this subsection. In the DRL performance optimizer, the heat pump is defined as an agent that can provide heat power while the environment is the BOPTEST test case emulator that can generate a new state.

4.2.1. Parallel training

In this paper, we develop an efficient parallel training scheme compatible with the Open-AI gym interface for BOPTEST [26] and the BOPTEST-Gym environment [37] to address the inefficiencies inherent in traditional sequential training methodologies. This scheme allows multiple DRL algorithms to train simultaneously in different simulated environments, substantially accelerating the training process.

The core of the developed parallel training scheme involves a dynamic port allocation mechanism. An automated dynamic modification of the YAML configuration files is designed to assign independent ports to each BOPTEST instance. This design enables the concurrent operation of multiple BOPTEST instances on the same physical or virtual machine without port conflicts. Furthermore, by leveraging Docker-compose and the modified YAML files, we facilitate the deployment of multiple BOPTEST instances in separate containers. Each container operates independently and runs parallel training tasks.

Another predominant feature of the scheme is the vectorization of multiple BOPTEST environments using OpenAI-Gym's SubprocVecEnv method. This allows a single DRL algorithm to train across multiple environment instances in parallel, significantly improving data throughput and training speed.

The parallel training scheme reduces the training time and improves the computational efficiency for DRL algorithms. This is particularly important for complex DRL models that demand extensive computational resources and time for training. Moreover, by optimizing resource usage, the scheme can enable conducting more experiments with limited computational resources, thus accelerating the innovation and validation process of control strategies. The developed parallel training scheme can be extended to other simulation environments and application scenarios deployed by Docker beyond BOPTEST.

The overall scheme is shown in Fig. 7 and the corresponding code has been merged into the master branch of the BOPTEST-Gym.[1]

4.2.2. Reward function

Similar to MPC, to achieve the dual aims of reducing thermal discomfort and operational expenses, the reward function is characterized by the negation of the incremental value of the objective function of MPC, as depicted in Eq. (12).

$$r_{k+1} = -(J_{k+1} - J_k), \tag{12}$$

[1] The link to the branch is https://github.com/ibpsa/project1-boptest-gym.

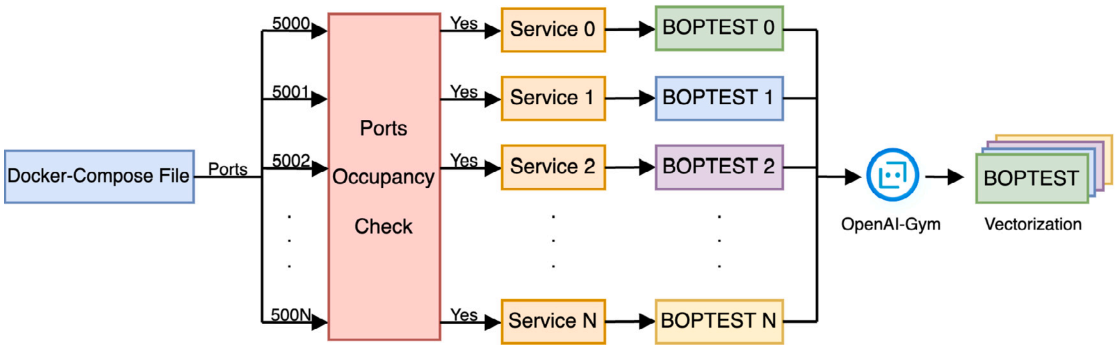


Fig. 7. Parallel training scheme.

Table 4
Observation variables.

Observation variable	Description	Min	Max
t [s]	Time of the week in seconds	0	60 480
T_z [K]	Zone operative temperature	280	310
T_{amb} [K]	Ambient temperature	265	303
$\dot{Q}_{\mathrm{rad}}$ [W/m^2]	Solar irradiation	0	862
$\dot{Q}_{\mathrm{occ}}$ [W/m^2]	Direct normal radiation	0	219
λ [1]	Electricity price	0.4	0.4
T_{lower} [K]	Lower bounds of the comfort zone	280	310
T_{upper} [K]	Upper bounds of the comfort zone	280	310

where J is specified in Eq. (11a). The agent will be trained to learn in a direction that simultaneously increases indoor thermal comfort and keeps energy consumption low.

4.2.3. Observation space

In this test case, indoor temperature and energy cost are influenced by multiple factors. The agent, during its training, cannot access information on all influencing factors, rendering the state space partially observable. To provide as much information as possible to assist the decision-making process of the agent, this study selects the observation variables as shown in Table 4. Additionally, it includes 6 h of historical measurement data and the forecast values for the upcoming 24 h. The observation values are normalized to a range of −1 to 1 to mitigate the impact of the original numerical scales. The observation space comprises entirely of continuous variables with a total dimension of 608.

4.2.4. Action space

For our specific test case, the action is the heat pump control signal, u_{hp}, with a range of [0,1]. Different DRL algorithms require distinct action spaces. Algorithms like deep deterministic policy gradient (DDPG), soft actor critic (SAC), advantage actor–critic (A2C), and twin delayed deep deterministic policy gradient (TD3) operate in a continuous action space. On the other hand, PPO and double deep Q learning (DDQN) algorithms function in a discrete action space. Therefore, we opt to divide the action into 10 equal intervals, $(0, 1/10, 2/10, \dots, 1)$, before passing it to the agent of PPO and DDQN. Consequently, the total dimension of the discrete action space is 11.

5. Results

The section presents and compares the simulation results of MPC, DRL and safe DRL in two different test periods: the peak heat day and the typical heat day in the BOPTEST. All algorithms are deployed on a local high performance computing (HPC) cluster with a Linux operating featuring dual-socket Intel(R) Xeon(R) Platinum 8338C CPU (32 cores per socket running at 2.6 GHz). The MPC is implemented using the do-mpc framework [38], while the DRL is implemented using the Stable Baselines3 framework [36].

Table 5
Safety metric comparison of MPC and PI controllers.

Controller	Scenario	r_{time}	r_{sev}	m_s
MPC	Peak heat day	0.012	0.004	0.016
PI	Peak heat day	0.093	0.003	0.096
MPC	Typical heat day	0.035	0.006	0.041
PI	Typical heat day	0.085	0.005	0.090

5.1. Model predictive control results

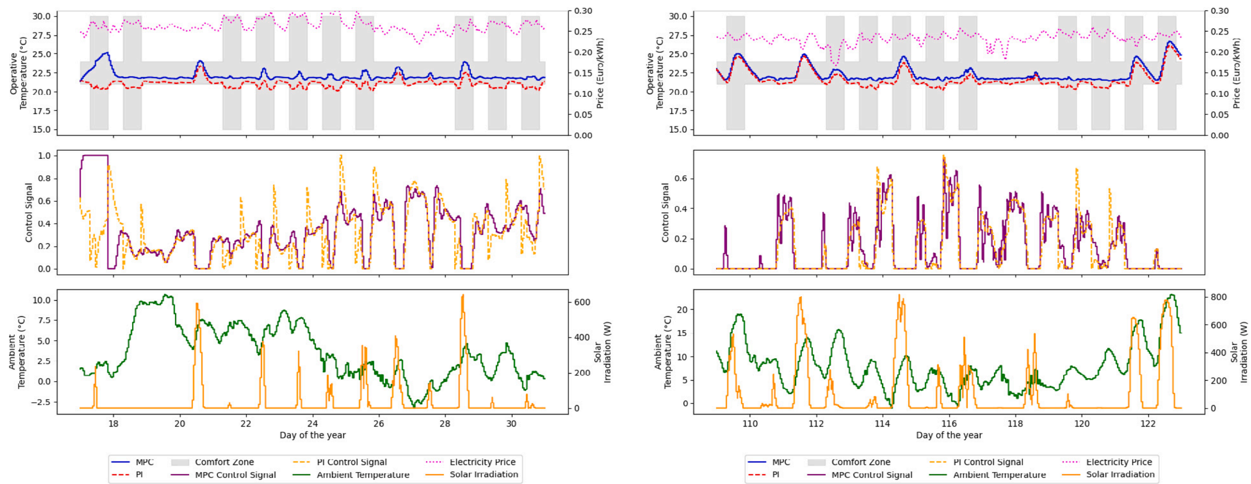
To validate the feasibility of MPC as a safety filter, this subsection compares the control results and safety performance of MPC with a rule-based baseline controller, a proportional–integral (PI) controller, during the testing periods.

In this subsection, the MPC is optimized with the IPOPT solver [39], with the results presented in Fig. 8.

The results demonstrate a comparative analysis of performance between the MPC and the rule-based, well-tuned baseline PI controller provided by the BOPTEST. It is observed that under the PI controller, the zone temperature is generally close to the lower constraint limit, and at times, it even exceeds the lower temperature limit. On the other hand, the MPC controller generally does not fall below the lower temperature constraint. During the peak heat day, the MPC system only approaches the upper temperature limit at a few points, while in the majority of instances, it remains within the upper and lower temperature boundaries. However, during several instances, particularly on typical heat day, it slightly exceeds the upper temperature limit due to elevated environmental temperatures and intense solar radiation.

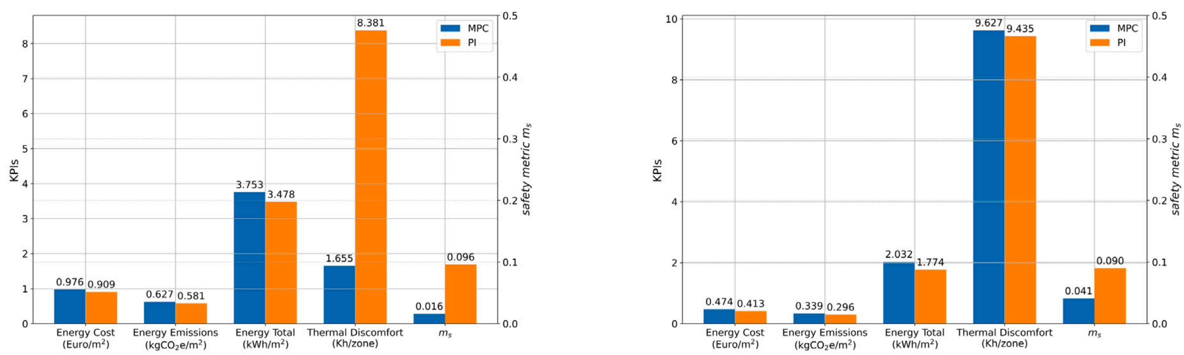
The comparison of the safety metric between MPC and PI controllers on the peak heat day and the typical heat day is illustrated in Table 5. The results indicate that under the test conditions of the peak heat day and the typical heat day, the unsafe time ratio r_{time} for the MPC system are 0.012 and 0.035, respectively. These figures are significantly lower than those for the PI system, which are 0.093 and 0.085, demonstrating the superior performance of the MPC in maintaining temperature stability. Considering extreme deviations, the MPC system exhibits maximum deviations (r_{sev}) of 0.004 and 0.006 during the test periods, compared to the PI controller's maximum deviations of 0.003 and 0.005. Although the PI controller shows slightly lower maximum deviations, the difference between the two controllers is minimal, at 0.001. This indicates that both systems maintain commendable performance even in the worst-case scenarios. Combining the aspects of time ratio and extreme deviation percentage, the overall safety metric (m_s) for the MPC and PI controllers on the peak heat day are 0.016 and 0.096, respectively, and 0.041 and 0.090 on the typical heat day, respectively. Based on those quantitative metrics, it is evident that the safety performance of the MPC system surpasses that of the PI controller.

BOPTEST further sets a range of standardized KPIs to facilitate an equitable assessment of performance. These KPIs encompass various



(a) Peak Heat Day (b) Typical Heat Day

Fig. 8. Simulation results of MPC and PI controllers.



(a) Peak Heat Day (b) Typical Heat Day

Fig. 9. KPIs of MPC and PI controllers.

parameters such as energy cost, energy use, CO_2 emissions, thermal discomfort, and computational time ratio. Fig. 9 showcases a comparison of these KPIs between the MPC and PI controllers during the peak heat day and the typical heat day scenarios. On the peak heat day, it is observed that the MPC results in higher energy cost, energy emissions, and total energy consumption than the PI controller, though the difference is marginal, less than 10%. The thermal comfort under MPC is 5.06 times greater than that achieved with PI control. This implies that MPC significantly enhances comfort levels with a relatively modest increase in energy expenditure. On the typical heat day, it can be found that due to the substantial increase in ambient temperature and solar irradiation intensity, the performance of both the PI controller and the MPC controller has declined to a certain extent, and both have experienced overshoot. While the PI controller exhibits marginally superior thermal comfort compared to the MPC controller, it exceeds the prescribed temperature boundaries more frequently over the entire period. The PI controller surpasses both the upper and lower temperature limits. In contrast, the MPC controller only exceeds the upper temperature limit during instances of peak solar radiation. Comparing the two test results, it can be stated that the MPC controller demonstrates overall superior performance over the PI controller regarding energy efficiency, thermal comfort, and safety.

The results above demonstrate that the MPC, although constructed from the linearized building energy system dynamics, satisfactorily adheres to safety constraints. Furthermore, its control effect is relatively stable, making it suitable as a safety filter to safeguard against potentially unsafe actions that may be made by DRL.

5.2. Deep reinforcement learning results

To ensure consistency with MPC, the control step size of the DRL algorithm is also set to 15 min. The other hyperparameters of the six DRL algorithms are kept at default values. The algorithms are trained excluding periods of peak heat day and typical heat day, and they are trained in dynamic electricity price scenarios. The testing is conducted under the highly dynamic electricity price scenario in BOPTEST.

The training processes of various DRL algorithms are illustrated in Fig. 10. We vectorize 32 BOPTEST environments for parallel training, with a step size of 15 min. Each episode covers two weeks, which is 43 008 steps, and each algorithm is trained for 10 million steps. Apart from TD3 and DDPG, the other four DRL algorithms, i.e., A2C, PPO, DDQN, and SAC, all converge to relatively high reward, demonstrating superior learning performance. As the reward values of these DRL algorithms are close, this paper chooses DDQN to show the efficacy of the safe DRL algorithm on the peak heat day and the typical heat day.

The performance of DDQN on the peak heat day and test heat day is shown in Fig. 11.

We can observe that it performs better on the peak heat day, mostly within the temperature constraint. On Test Heat Day, there is a significant overshoot due to solar irradiation and increased ambient temperature. Compared with MPC, DDQN make more bold attempts, which sometimes lead to performance improvements and sometimes lead to performance drops.

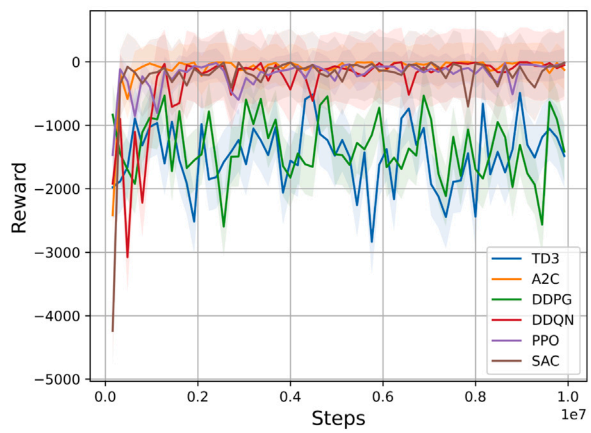


Fig. 10. Training process of DRL algorithms.

Further visualization of KPIs including the safety metric is shown in Fig. 12. It can be seen that on the peak heat day, DDQN achieves a thermal discomfort level of only 0.523 and a safety metric of 0.015, which is lower than the MPC and PI controllers analyzed earlier. But at the same time, its energy cost and total energy consumption have also increased, reaching 1.007 and 3.829 respectively, ranking the highest among all controllers. On the typical heat day, on the contrary, the thermal comfort decreases rapidly with thermal discomfort value reaches 30.202, and the safety metric is ten times higher than peak heat day, while its energy cost and consumption are about 2.3% higher than MPC. The results above show that the results of DDQN are unstable and further improvement of its overall performance is needed.

5.3. Safe deep reinforcement learning results

This subsection tests the proposed safe DRL algorithm to harnesses the exploratory prowess of DRL for performance enhancement while utilizing the predictive strengths of MPC to dynamically safeguard the DRL action at each step.

The simulation results of safe DRL are shown in Fig. 13.

In comparison to the performance of the DDQN during the test periods depicted in Fig. 11, it is observed that safe DRL leads to a more stable trend in the changes of operative temperature due to the integration of MPC as a safety filter. This combination notably diminishes both the overshoot and undershoot.

Fig. 14 further illustrates the results of the KPIs and the safety metric m_s for safe DRL during the testing period. It is observed that on the peak heat days, safe DRL, benefiting from the safety filtering function of MPC, not only reduces the m_s and thermal discomfort to zero but also shows a decrease in energy consumption and carbon emissions. These indicators outperform those of DDQN, MPC, and PI controllers, demonstrating a substantial improvement in performance. On the typical heat days, despite the high environmental temperature and solar radiation causing some overshoot, safe DRL still outperforms DDQN, MPC, and PI controllers in terms of KPIs, delivering more satisfactory results in the context of building energy management. During the testing phase, the maximum computation time required by MPC at each step is recorded as 2.57 s, demonstrating that significant performance improvements are achieved with a relatively small investment in time.

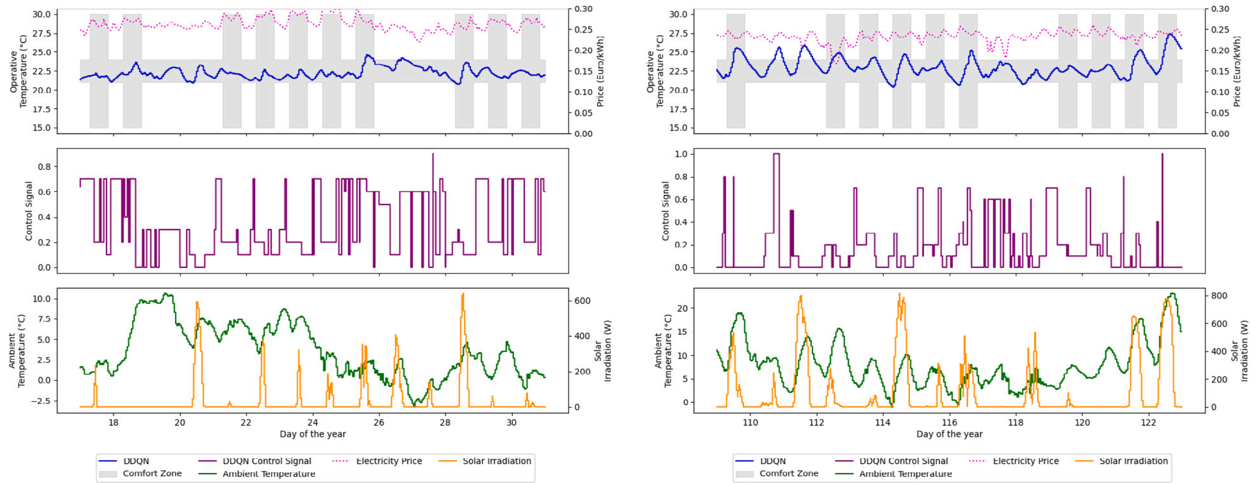
The metrics comparison of algorithms is shown in Table 6. The results reveal that during peak heating days, safe DRL achieves the highest thermal comfort and best safety performance, albeit at the cost of an acceptable higher energy use and CO_2 emissions than PI controller. On typical heating days, safe DRL outperforms other algorithms in terms of KPIs, but the safety performance is slightly weaker than MPC. It can be seen that with the synergy of MPC, safe DRL reduces the safety risk of DDQN. On typical heating days, it achieves a level of safety performance comparable to that of MPC. During peak heating periods, safe DRL attains superior safety performance with the proposed safety metric m_s equals to zero, indicating that at certain time steps, the actions provided by MPC are not globally optimal. Due to the exploratory nature of DRL, more optimal actions are identified during these steps, and the MPC filter successfully retains these actions, thereby enabling safe DRL to surpass the safety performance of MPC. This observation validates the effectiveness of the MPC filter and the potential of exploration-based DRL to generate more optimal actions.

To further compare the DDQN and safe DRL strategies and quantify the frequency of MPC intervention, Fig. 15 shows the comparison of control signals of DDQN and safe DRL. The visualization reveals key distinctions between safe DRL and DDQN control signals, highlighting instances where safe DRL defers to MPC, denoted by the presence of orange markers.

In Fig. 15(a), it is noted that MPC and DRL execute actions 686 and 658 times, respectively. On the 26th and 27th days, there is a noticeable increase in interventions by MPC. This response aligns with the data in Fig. 13(a) when the temperature curve is approaching its upper limit at these times. Consequently, MPC predicts the risk of overshooting in the future and filtered the actions of DRL at these moments to ensure the system's safety performance.

As depicted Figure in 15(b), on the typical heat day, MPC intervene for 950 time steps. This indicates that, according to assessment of MPC, DDQN is more prone to making unsafe actions on the typical heat day compared to the peak heat day. Taking the period from day 109 to day 111 as an example, it is clear that compared to the control actions of DDQN, MPC's intervention significantly reduced the control actions during these times to prevent the overshooting that DDQN would have likely caused (as shown in Fig. 11(b)). Despite the challenges posed by intense radiation and high ambient temperatures, which prevented the indoor temperature from being completely safeguarded within the safe range, a notable improvement in safety is observed when compared to the DDQN in the typical heat day results. This demonstrates the efficacy of the MPC safety filter in enhancing system safety under challenging environmental conditions.

The analysis indicates that the proposed safe DRL framework has been successful to balance the performance optimization and safety ensurance. It incorporates MPC as a safety filter, thereby enhancing the KPIs and safety performance of the DRL optimizer in the safe DRL framework. It effectively predicts potential unsafe states and filters out potentially unsafe actions determined from the DRL optimizer. The safe DRL implementation may imply its broad applicability in enhancing safety across various applications beyond the specific test case. The use of a first-order dynamic model in the MPC safety filter in this context illustrates its potential adaptability in other scenarios where we can construct MPC as a safety filter based on first-order system dynamics equations derived from physical laws. Additionally, we can model the target building's energy environment using the BOPTEST emulator and employ the proposed parallel training algorithm to accelerate the training process of the DRL algorithm. Note that the DRL training process in the safe DRL framework does not involve complex parameter tuning, thereby suitable for integrating off-the-shelf DRL tools and easier for engineers to use. Finally, using sim2real technology, the algorithm can be deployed in real-world scenarios. However, we also need to notice that the false cues received from the simulation environment could mislead the agent leading to poor behavior during testing, as pointed out in [25]. To further mitigate the risks associated with practical implementation, techniques such as gradual transition or maintaining a human-in-the-loop approach during the initial stages of execution can be employed.



(a) Peak Heat Day (b) Typical Heat Day

Fig. 11. Simulation results of DDQN.

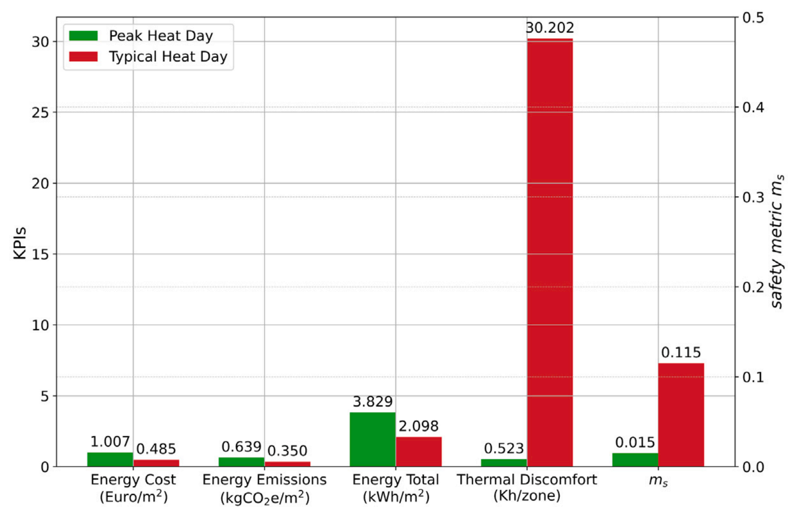
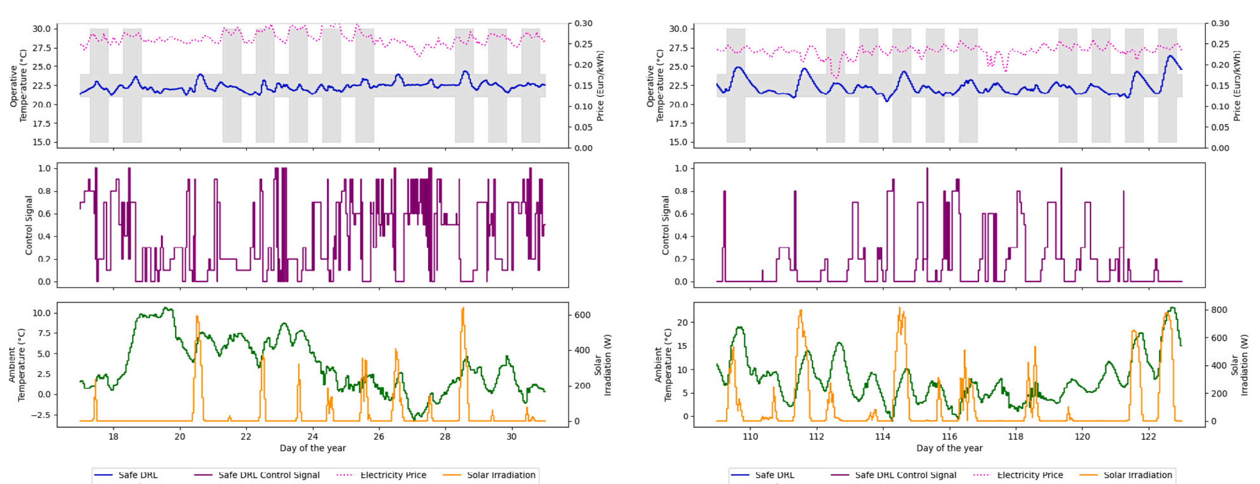


Fig. 12. KPIs and safety metric of DDQN.



(a) Peak Heat Day (b) Typical Heat Day

Fig. 13. Simulation results of safe DRL controller.

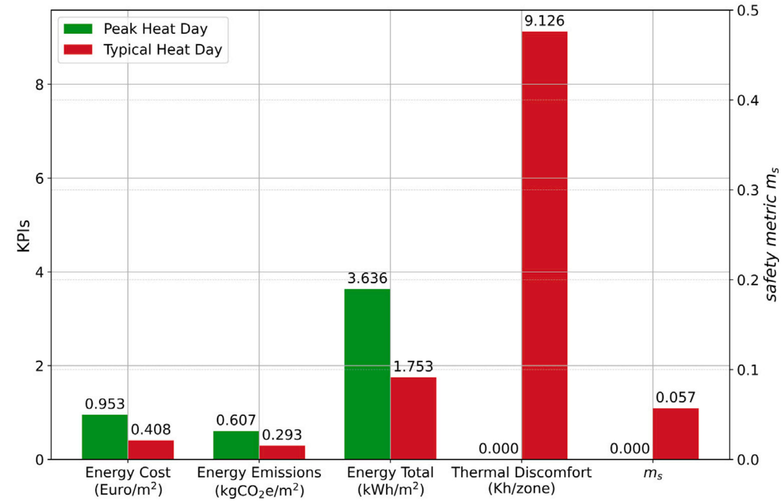


Fig. 14. KPIs and safety metric of safe DRL.

Table 6
Metrics comparison of algorithms.

Algorithms	Metrics									
	Energy cost (Euro/m^2)	Energy emissions (kgCO2e/m^2)	Energy total (kWh/m^2)	Thermal discomfort (Kh/zone)	Safety metric	Energy cost (Euro/m^2)	Energy emissions (kgCO2e/m^2)	Energy total (kWh/m^2)	Thermal discomfort (Kh/zone)	Safety metric
	Peak heat day					Typical heat day				
PI	**0.909**	**0.581**	**3.478**	8.381	0.096	0.413	0.296	1.774	9.435	0.09
MPC	0.976	0.627	3.753	1.655	0.016	0.474	0.339	2.032	9.627	**0.041**
DDQN	1.007	0.639	3.829	0.532	0.015	0.485	0.35	2.098	30.202	0.115
Safe DRL (with DDQN)	0.953	0.607	3.636	**0.000**	**0.000**	**0.408**	**0.293**	**1.753**	**9.126**	0.057

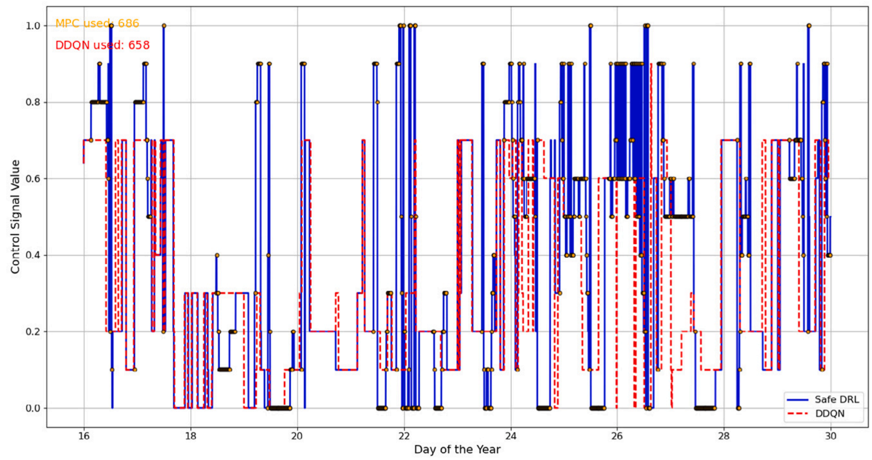
6. Conclusion

This paper proposed a novel safe deep reinforcement learning (DRL) framework and tested it at different testing periods in the BOPTEST environment. The safe DRL algorithm demonstrated better security and robustness without performance degradation. The main conclusions of this work were as follows:

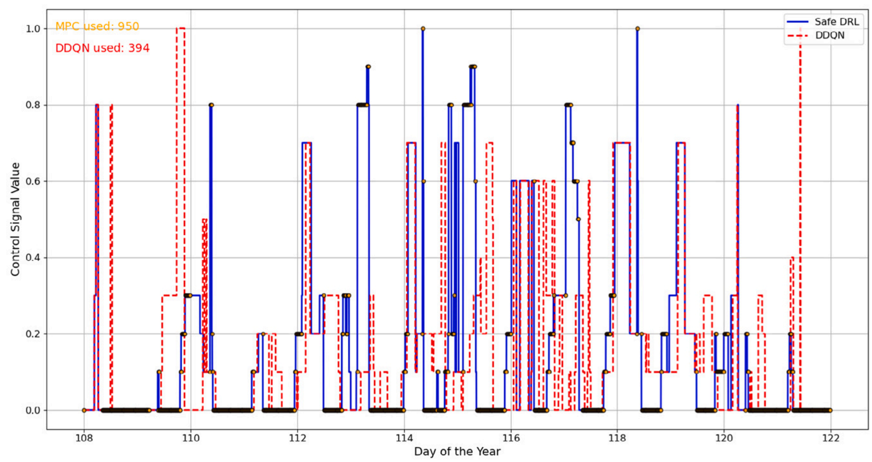
- The paper proposed a safe DRL algorithm, using DRL to explore the optimality in uncertain environments and model predictive control (MPC) as a safety filter to check DRL actions at each step. The algorithm eliminated the need for the effort-extensive tuning of DRL algorithms and allowed the use of off-the-shelf DRL tools, which made it easier for engineering implementation. The results demonstrated that using a maximum of 2.67 s per step can significantly enhance the robustness of DRL control actions, greatly improving the overall KPIs and the safety performance of the control system.
- This paper developed a parallel training scheme for BOPTEST, employing a dynamic port check and allocation mechanism. In the parallel training scheme, multiple BOPTEST environments were vectorized through OpenAI-gym interface to accelerate the training and testing process of DRL. Computational resources were efficiently utilized to compare the effectiveness of six mainstream DRL algorithms with the scheme. After a comparative analysis, DDQN was selected for the safe DRL execution. This methodology might be extended to other simulation environments and application scenarios that were deployed using Docker.
- A safety metric was proposed to assess the safety performance of algorithms during testing periods. Both the severity and the duration of unsafety were considered in the metric including the percentage of maximum overshoot or undershoot, and the proportion of total time exceeding constraints. Comparative results showed that the safety metric could well capture the safe and unsafe behaviors of the tested system and serve as a satisfactory indicator of the safety performance of the safe DRL.

There are several important future research directions:

- The performance of the proposed approach could be tested on larger HVAC systems with multiple control inputs.
- We also note that, while incorporating heat flux meters into the setup is technically feasible, they are not widely used today. Therefore, future research could explore the economic viability of using heat flux meters, or focus on innovations in sensor technology that might provide less invasive and more cost-effective methods for continuous monitoring of heat flux.
- The accuracy of the MPC model affects its performance, and consequently, its effectiveness as a safety filter. The MPC setup in this study did not consider uncertainties such as noise and bias. Future research could explore the validation of MPC variants like robust MPC and their performance as safety filters.
- The results of this paper demonstrate the promising potential of safe DRL in building energy systems. Future work may include exploring its applicability to other control objectives such as indoor humidity.



(a) Peak Heat Day



(b) Typical Heat Day

Fig. 15. Comparison of control signals of DDQN and safe DRL.

CRediT authorship contribution statement

Xiangwei Wang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Peng Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Renke Huang:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Xiuli Zhu:** Writing – review & editing, Supervision, Investigation. **Javier Arroyo:** Supervision, Resources. **Ning Li:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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